Status	Finished
Started	Sunday, 18 October 2026, 12:30 AM
Completed	Sunday, 18 October 2026, 12:36 AM
Duration	5 mins 53 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The search continues in the left or right sublist ✓
- b. The search continues from the beginning
- c. The search stops
- d. The list is sorted

Question **2**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

- a. When the item is the last element in the array
- b. When the item is not the array at all
- c. When the item is somewhere in the middle of the array ✓
- d. Item is the last element in the array or item is not there at all

Question **3**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 99

b. 90 and 99 ✓

c. 90 and 94

d. 89 and 94

Question **4**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n ✓

b. n/2

c. log n

d. n-1

Question **5**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Binary search

b. Hash search

c. Linear search ✓

d. Interpolation search

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Dividing the list in half repeatedly
b. Checking each element sequentially ✓
c. Sorting the list before searching
d. Skipping every second element

Question **7**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search has a lower time complexity on large, sorted lists ✓
b. Binary search does not require dividing the list
c. Binary search can find multiple instances of the target element
d. Binary search works on unsorted lists

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Hashing
b. Linear search ✓
c. Dynamic search
d. Binary search

Question **9**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Sort the list
b. Compare the target element with the first element in the list
c. Divide the list into two equal parts
d. Compare the target element with the middle element in the list ✓

Question **10**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. log n
b. n ✓
c. n/2
d. 1

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search requires the list to be sorted.
b. Linear search is more efficient than binary search on large lists.
c. Linear search can be applied to both sorted and unsorted lists. ✓
d. Linear search divides the list into halves.

Question **12**

Correct

Mark 1.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The search stops
b. The higher middle element is chosen as the middle element ✓
c. The middle element is chosen randomly
d. The lower middle element is chosen as the middle element

Question **13**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By dividing the list into halves
b. By using a hash function
c. By sorting the list first
d. By comparing each element in the list sequentially ✓

Question **14**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Jump search
b. Hashing
c. Binary search
d. Linear search ✓

Question **15**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the entire list

b. In the right sublist

c. At the beginning of the list

d. In the left sublist ✓